

B

Operationalized Statements

Table B.2: Data Owner: Functional and Problem Statements

Key	NFR	Functional Statement	Problem Statement
NFR1	Accessibility	I need to access my identity data whenever I want.	Limited access to my identity data when I need it would be acceptable to me.
NFR2	Authenticity	I want my credentials to carry a signature that proves they are genuine and untouched.	If I had to use credentials that might be fake or altered, I could cope with it.
NFR3	Autonomy	I want to manage my digital identity myself, without relying on a central authority.	If I had to ask someone else to change or share my digital identity, it would be fine for me.
NFR4	Availability	My digital identity should be available whenever I need it, even if some services are down.	Being prevented from using my credentials due to a service outage would be acceptable for me.
NFR5	Compatibility	I like it when my digital credentials work smoothly across different websites and apps.	I could handle my digital credentials working in some places but failing in others.
NFR6	Consent	I want services to ask for my permission each time they use my identity information.	If my identity information were used without my permission, it would be manageable for me.
NFR7	Control	I want to choose exactly which parts of my digital identity I show or hide.	If I lacked control over which identity details were shared, it would be acceptable to me.
NFR8	Cost	I prefer identity solutions that let me store and use my credentials without extra fees.	I could tolerate paying high fees to keep or present my credentials.
NFR9	Decentralization	I want to create and use new digital identities without depending on any single company or server.	I would be fine relying on one provider to store or verify my identity.

Continued on the next page

Table B.2: Data Owner: Functional and Problem Statements (continued)

Key	NFR	Functional Statement	Problem Statement
NFR10	Existence	I want to be able to create a new digital identity whenever I choose, without asking anyone.	It would be fine for me if I needed approval from a third party before creating a new digital identity.
NFR11	Interoperability	I want my digital credentials to work on any website or app, no matter which platform or provider the website or app uses.	Having credentials that work in one place but fail on other services would be acceptable to me.
NFR12	Persistence	I want my digital-identity credentials to stay valid and usable for as long as I need them.	If my credentials expired or disappeared before I chose to remove them, that would be acceptable for me.
NFR13	Portability	I want to move my digital credentials to another phone or wallet whenever I switch devices.	It would be manageable for me if my credentials failed to transfer to a new device.
NFR14	Privacy	I want to share only the minimum details about myself when I prove my identity.	If I had to reveal more personal information than necessary, I would be fine.
NFR15	Protection	I need my identity data to be protected from unauthorized access or tampering during storage and transmission.	If someone could read or change my identity information without permission, I would accept that.
NFR16	Recoverability	I want to be able to restore my digital identity if I lose my phone or keys.	If I were unable to recover my identification after losing my device or keys, it would be manageable for me.
NFR17	Representation	I want separate digital identities for different situations.	I would be comfortable using a single digital identity in every situation.
NFR18	Security	I need assurance that both the technology and its operators secure my identity data against threats.	I could handle the possibility of my identity data being exposed in a breach.
NFR19	Single Source	I want to have one reliable place that holds the official version of my identity.	If different services had conflicting versions of my identity, I would be okay with that.
NFR20	Standard	I prefer credentials that follow common standards so every service can read them.	I would accept credentials that violate standards and fail various services.
NFR21	Transparency	I want to see who has accessed or shared my identity information and when it happened.	I could tolerate the uncertainty of which services have utilized my identifying information.
NFR22	Usability	Using my digital identity should be quick and straightforward.	Having to go through many steps to use my digital identity would be acceptable to me.
NFR23	User Experience	I want managing my digital identity to feel simple and intuitive.	I would find it acceptable to use a confusing or complicated identity-management app.
NFR24	Verifiability	I want the people or services I interact with to check my digital credentials instantly, without asking me for extra proof.	A delay in the verification of my credentials would be tolerable.

Table B.3: Verifier: Functional and Problem Statements

Key	NFR	Functional Statement	Problem Statement
NFR1	Accessibility	I need to access issuer information instantly whenever I verify a credential.	I can verify a credential effectively without retrieving issuer details.
NFR2	Authenticity	I must be able to prove that every credential I check is genuine and unaltered.	I could accept credentials without verifying their authenticity.
NFR5	Compatibility	I need our service to accept and check credentials that come in different formats or protocols.	Rejecting a credential just because it was issued under another standard would be manageable for us.
NFR6	Consent	Before using someone's identity data, I want to receive clear consent from the holder every time.	Processing identity data without explicit consent from the holder would be fine for my organization.
NFR8	Cost	I need to verify credentials without incurring high transaction or storage fees.	I am comfortable paying high costs to verify credentials.
NFR9	Decentralization	I need to verify credentials without relying on any single service.	I am comfortable relying on a single provider to check credentials.
NFR11	Interoperability	I need to verify credentials that come from many different platforms or blockchains.	Accepting credentials from only one ecosystem is sufficient for our service.
NFR14	Privacy	When I check someone's identity, I want to see only the data that is strictly needed for my service.	Requesting more personal data than necessary would be acceptable for our service.
NFR15	Protection	I need to verify identity data over a channel that keeps it safe from misuse or tampering.	I would maintain trust in the system even if identity data could be intercepted or altered during verification.
NFR18	Security	I need confidence that security threats to identity data are effectively managed during verification.	Effective security management during verification feels optional to me.
NFR20	Standard	I prefer to accept only credentials that follow recognized standards and schemas.	I would be fine accepting credentials that fail to meet the required standards and schemas.
NFR21	Transparency	I need to see a clear, unalterable record of how and when credentials are verified.	An audit trail of credential checks feels nonessential to me.
NFR24	Verifiability	I need to confirm on my own that any credential I receive is genuine and still valid.	I trust credentials regardless of whether I can verify their validity myself.

Table B.4: Issuer: Functional and Problem Statements

Key	NFR	Functional Statement	Problem Statement
NFR1	Accessibility	The ability to digitally sign each credential I issue to guarantee its authenticity and integrity is important to me.	Issuing credentials without a verifiable digital signature would be acceptable to me.
NFR5	Compatibility	I need to issue credentials that work across both modern identity platforms and traditional identity systems.	Credentials that are incompatible with existing legacy systems would be acceptable.
NFR6	Consent	I need to obtain explicit permission from the holder before issuing any credentials about them.	Issuing a credential without the holder's clear consent would be tolerable.
NFR8	Cost	I need to issue credentials without incurring high transaction or storage fees.	A system that charges high fees for issuing credentials would be a minor factor in my decision to use it.
NFR9	Decentralization	I need to issue and revoke credentials using a decentralized registry without relying on any single central authority.	I feel comfortable issuing credentials that rely on a single central provider.
NFR11	Interoperability	I want the credentials I issue to work with any system or service, no matter the platform.	It would be manageable if some services rejected the credentials I issued.
NFR14	Privacy	I need to include only the personal data strictly necessary when issuing a credential.	Including any unnecessary personal data in a credential would be acceptable.
NFR15	Protection	I need identity data to be safeguarded from misuse during the credential-issuance process.	The possibility of credentials I issue being intercepted or tampered with would be an acceptable risk.
NFR18	Security	I must ensure that identity data is secure against threats during credential issuance.	Security measures during credential issuance feel nonessential to me.
NFR20	Standard	I need to define and publish my credential schemas in widely accepted formats so that any system can process the credentials I issue.	I could tolerate credentials with non-standard or outdated schemas being unusable by many services.
NFR21	Transparency	I need to record each credential I issue on a public, tamper-proof ledger so anyone can audit my actions.	A lack of a tamper-proof record for the credentials I issue would be an acceptable situation.
NFR24	Verifiability	I must digitally sign each credential and publish its public key or status so it can be independently verified.	I would remain confident in the credentials I issue, even if they lack a verifiable signature or current status.